

UWB Based Indoor Localization Techniques

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Abstract— Ultra wideband (UWB) is a technology for transmission of the data or information using the techniques which cause the spreading of radio energy over a wide frequency band (3.1-10.6GHz) with a very low power spectral density. Localization is the technique which is commonly employed in wireless network to improve routing and enhance the security. It is the method of determining the position of the object, people, and equipment. Indoor positioning system (IPS) has many applications like providing the indoor navigation system for blind, visually impaired people, location of the devices inside the building, locating the emergency exit in the smoky area, tracking the kids in the crowded areas, tracking the expensive equipments etc. UWB localization techniques have many applications in different fields, such as Robotics, Body area network (BAN), Personal area network (PAN), Surveillance area (mining area, military area, navigation area, underground tunnel etc). This paper contains the survey of different localization techniques such as Time of arrival (TOA), Time differential of arrival (TDOA), Received signal strength (RSS), Angle of arrival (AOA) method. The SWOT (Strength, Weakness, Opportunities, threats) analysis has described to analyze the UWB positioning technologies. In Wireless sensor Network (WSN), it gives robust performance in dense multipath environment and creates the best choice for indoor localization.

Keywords-Ultra Wide Band, Localization, Positioning, TOA, TDOA, RSS, AOA, SWOT, Indoor positioning, WSN

I. INTRODUCTION

Now a days the utilization of wireless communication increases rapidly, so the requirement of high data rate also increases, for this there is a need of more data capacity to satisfy the user's demands. The high data rate communications can be achieved with increasing the communication signal bandwidth. Ultra wideband (UWB) is a digital pulse wireless technology for transmission of digital data with a frequency range of 3.1GHz to 10.6GHz. It requires low power (approximately 0.5mw) for short or small distance (approximately up to 15m) of operation. The limited power spectral density (PSD) of -41.3dbm/MHz, allows it to be easily and safely deployed with existing wireless communication system without causing any interference [1]. UWB broadcasts the pulses on through the doors and different obstacles will again reflect the signal with more limited carrier signal through a wide spectrum having number of frequency channels. It causes no interference and the regulatory bodies are moving forward cautiously, hence the users who already have spectrum allocation are not affected. The high rate communication with low energy consumption and multipath resistant

property of UWB allows for wireless ranging application with high accuracy [2]. UWB is a technology for transmission of information over a wide range of frequency band with the lower power spectral density. Comparing the conventional radio transmission and UWB we find the former transmits information with varying power level, frequency level and the phase of the sinusoidal wave, the later transmits the data by generating the radio energy by occupying a larger bandwidth. Pulses are very short: approximately less than 60cm for 500MHz and less than 23cm for 1.3GHz. Hence there is no overlapping of reflected signals with the original signal occurs [3]. Localization is the technique which is commonly employed in wireless network to improve routing and enhance security. It is also defined as the process of determining the position of people, object, and equipment. The Positioning is categorized into two types: Indoor positioning and Outdoor Positioning. The outdoor positioning is conducted outside the building area and the indoor positioning is performed inside building area like hospitals, house, malls, university campus etc. Now a day's most of the applications requirements are based on real-time tracking inside the buildings accurately, hence indoor localization service has become an important prerequisites in the market [4].

II. INDOOR POSITIONING TECHNOLOGY

Depending on infrastructure the indoor positioning technology divided into two categories: building dependent and building independent as shown in Fig.1. The building dependent again categorized into building infrastructure based and dedicated infrastructure based positioning Technology [5][6].

The details of technologies are given below.

RFID (Radio frequency identification)-

This technology is based on exchanging the different frequencies of radio signal between the main components: reader and tag. Tag emits radio signal that will receive by reader and viceversa. Both of them use predefined frequencies, protocol to send and receive the data. The tag consists of microchip having storing capacity 2kb of data and a radio antenna. The Reader consists processor, that interface to connect to a server. It also associated with transceiver, power supply, antenna. The methods that are used with RFID like proximity, Received signal strength (RSS).
UWB-

The FCC defines UWB as Radio frequency signal, that occupies more than 20% of central carrier frequency having frequency range more than 500MHz. This allows the UWB transmitter to spread the large amount of data with consumption of less amount of energy. It uses the TDOA method for finding out the position of the target.

Infrared (IR)-

It uses the invisible spectrum of light. This technique involves LOS communication up to 16Mbps. The Diffuse Infrared is more stronger than the direct Infrared with longer range 9m-12m. Proximity and angle of arrival method is frequently used in this IR technology.

Ultrasonic-

It is defined as the mechanical wave due to the oscillation of the pressure that is transmitted by the medium. The relative distance can be estimated with TOA using ultrasound pulses travelling from emitter to receiver.

Zigbee-

Zigbee provides the security of the network along with support on the top of IEEE.802.15.4 Specifications. It covers short distance with less data rate personal area network. It has small node. It has low cost with lower complexity. It is designed for low data throughput with lower power consumption. It creates location by coordinating and communicating with neighboring nodes.

Wireless Local Area Network (WLAN)-

It uses the carrier sense multiple access collision avoidance technique. This standard known as local wireless networking standard. It uses the RSS positioning method and extracts the 802.11 network to run on-off the shelf WLAN hardware. The positioning methods like TOA, TDOA, and AOA are used very less due to its angular measurement and time delay complexity.

Cellular Based-

The Global system for mobile communication (GSM) can be used in WLAN with the less location accuracy. It operates in a licensed band and immune to the interference in same region of frequencies. There is a possibility of using the positioning method in this network in a building if it is covered with strong RSS. The fingerprinting method is used in GSM which is based on RSS.

Bluetooth-

It is used for low power with peer to peer communication. It is meant for WPANs. It operates on 2.4GHz ISM band. It uses the proximity and RSS method for position estimation [7][8].

Dead Reckoning-

In this method object can find its location by finding the past position and velocity. It uses the different method for improving the accuracy and to nullify the error. Pedestrian dead reckoning is used for estimation of length of step along with the direction of the walking person.

Image Based Technology- It is also known as optical method which include camera and computer based vision technology. This technology depends on different factors like advanced technology detection, rate of data transmission, image processing algorithms etc. Again it is divided into two categories: static sensor, which locate the moving object in the image and ego motion system.

Pseudolites-

These are used to generate the GPS-like signals for indoor environment for receiving the signal from transmitters

instead of satellites because satellite signals do not penetrate through the different obstacles like building, coal mines, trees, tunnels etc. For indoor environment it produce negative effect due to signal multipath, interference, weak time synchronization due to less accurate clock, carrier phase ambiguities [9][10].

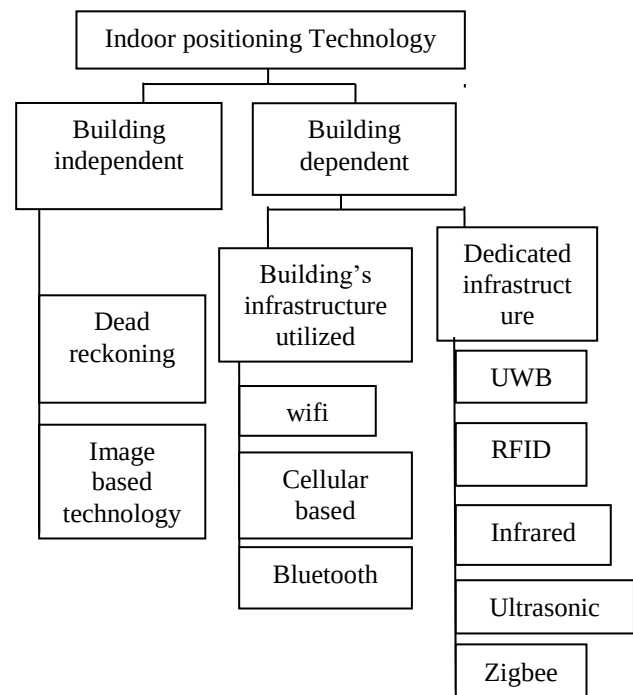


Fig.1. Classification of Indoor positioning Technology

III. LOCALIZATION DETECTION TECHNIQUES

The different location detection techniques are proximity type, triangulation type. Proximity is the simplest method of connectivity but it produces less accuracy. The triangulation type is direction based (angle based) and distance based. The triangulation uses the geometric property of the triangle to find the location of the target. It is again two type's lateration and angulation. The different techniques that are based on propagation time are TOA, TDOA, and RTOF. The lateration method depends on propagation time [11].

Angle based method-(Angle of Arrival-AOA)

It estimates the angle of arrival of mobile signal from the known location to multiple base stations. It uses the directional antenna or antenna arrays for finding the direction. It uses the geometric relationship to determine the location of intersection of angle for each signal source bearings from known reference point which is shown in Fig.2. The limitation of this method is, it uses the additional antenna with increased cost of the AOA system. It is also affected by multipath and non line of site propagation of signal. So it is not good for indoor Positioning, for this it changes the direction of arrival and degrades the accuracy of the system. AOA estimation algorithm have higher complexity due to the antenna array geometry [12].

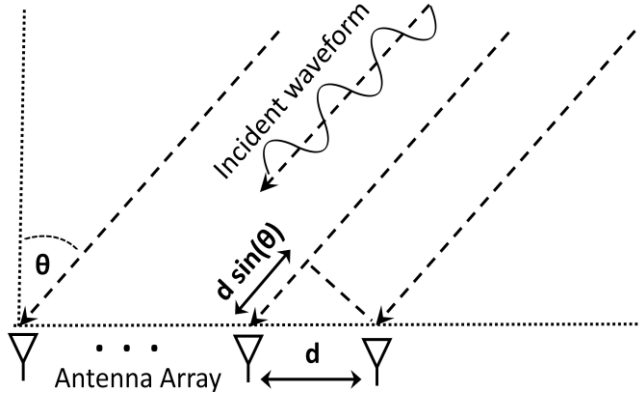


Fig.2.AOA based Localization

Time based method-

It includes lateration, trilateration and multilateration method. It finds the position of an object by measuring the distance from different reference points.

TOA/TOF (Time of Flight) –

This system is based on accurate synchronization of arrival time of signal from mobile device to several receiving beacons. After receiving the signal, the distance is calculated from transmission time delay and speed of the signal. The TOF value multiplied by speed of light $c=3*10^8$ m/second, determines the physical distance between Transmitter and Receiver.

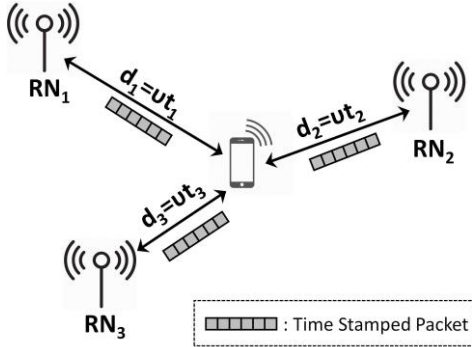


Fig.3.TOA Localization

In the given fig.3, three different nodes are used to find the distance between the reference nodes and the device. TOF requires the synchronization between transmitter and receiver, but the limitation is it cannot eliminate the localization error in non line of site between the Transmitter(Tx) and Receiver(Rx), because the obstacles defect the emitted signals and the signal traverse the longer path, which causes the increase of propagation time between the Tx and Rx. considering T1 be the time when Tx A sends a message to Rx B and that receives at time T2, where $T2=T1+Tp$, where Tp is the signal traverse period from Tx to Rx[13]. Now the distance is

$$D_{AB} = (T2-T1) * V,$$

Where V is the signal velocity.

Time Difference of Arrival (TDOA)-

It is based on the difference of time of arrival of the signal sent by an object and received by the receivers, which are

shown in Fig.4. By this the location of the object is determined [14].

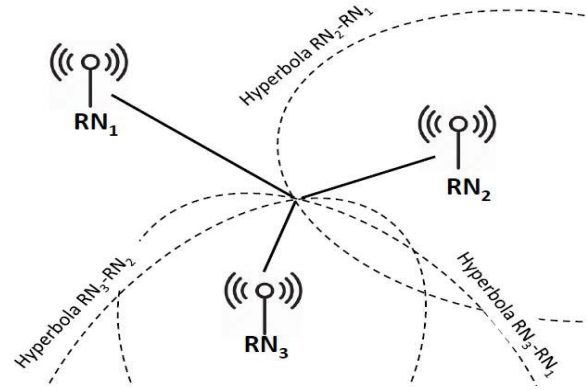


Fig.4.TDOA based Localization

Here the TDOA measurements are converted into physical distance. Hence

$LD_{AB} = C * TD_{AB}$, where TD_{AB} is the propagation time from Tx A and B and LD_{AB} is the physical distance value. Now the Rx is located on the hyperboloid as shown in the fig.4. Hence

$LD_{AB} = \sqrt{[(X_A-x)^2 + (Y_A-y)^2 + (Z_A-z)^2] - [(X_B-x)^2 + (Y_B-y)^2 + (Z_B-z)^2]}$, where the (X_A, Y_A, Z_A) are the coordinates of Tx or Reference node A and (x, y, z) are the coordinate of the Rx or user. To find the exact location of the Rx, TDOA requires at least three Tx, since intersection of three or more forms the hyperboloid. The accuracy depends on the bandwidth of the signal, sampling rate at the Rx and direct line of site between the Tx and Rx. Here the strict synchronization is only required between the Tx's, but in TOF method it required between the Tx and Rx[15].

Received Signal Strength Indicator (RSSI)-

In this method the tracked target measures the signal strength for received signal from different transmitters to find the distance between the transmitter and receivers. By this receiver is able to find its position with respect to the transmitter's node. The signal power strength is measured in decibel watts (dBw) or miliwatts (mw). The higher is the RSS value, the lower is the distance between Transmitter and Receiver. Using the RSS and path loss propagation model, the distance between the Tx and Rx can be calculated using the equation:

$$RSSI = -10 \log_{10}(d) + A,$$

Where n is the path loss exponent.

$n=2$ in free space,

$n=4$ in indoor environment.

A is the RSS value at reference distance from the receiver. The RSS based technique is of two types: Trilateration and fingerprinting[16]. In the first one it uses the RSS measurement to find the distances with three different reference nodes for estimation of current node. In the second one it requires the collection of datasets of RSS fingerprints, which is matched with the online measurements with the closest fingerprints in the dataset to determine the location.

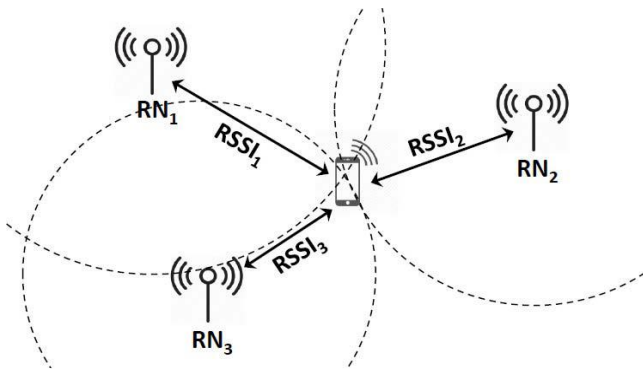


Fig.5.RSSI based Localization

This technique is relatively chip to implement and obtained by low complexity algorithm. It does not require the expensive time synchronization and no clock bias. The RSS based approach is simplest and cost efficient. In non line of site condition it suffers from poor localization accuracy due to the signal attenuation and different obstacles. The fluctuation also occurs due to the multipath fading and indoor noise[17].The accuracy error can be reduced by using averaging mechanisms or by using different filters.

IV. SWOT ANALYSIS

The aim of this analysis is to identify the internal key factor (strength, weakness), external key factor (opportunity, threat) that will affect the target.

The key factors include:

Strength-

Due to the less power consumption of signal, it does not overlap with other existing radio system. It has large bandwidth, hence, it provides immune to multipath effect. This high bandwidth offers an improved capacity of the channel with high data rate. According to the Shannon's law, the capacity of the channel:

$$C = B \log_2 (1 + S/N)$$

The larger the PG, the higher the immunity distortion and noise. It has the greater penetration to obstacles than other conventional signals. Its short pulses characteristic provides resolvability of multipath component. Its carrier less transmission property offers a simple and small hardware complexity. It does not require power amplifier due to its low power consumption [18].

Weaknesses-

There may be possible condition for interference with the nearby system which operates in the UWB spectrum due to the misconfiguration, because UWB frequency range of communication is 3.1GHz to 10.6GHz. There may be Interference which can be avoided by minimum mean square error (MMSE) multiuser detection scheme to nullify the interference. It uses very short pulse, which may take a long time for synchronization [19].

Opportunities-

Now a day's UWB has become the choice for tracking systems due to its high accuracy. It is also used in medical applications for its sub millimeters accuracy. For indoor localization system, it provides better time resolution. Its application also found in industrial water house, shipboard environment, military, noisy environment [20].

Threats-

Generally it has a very less negative impact since it avoids the interference, but it is slightly expensive than other systems. When the link distance to the height of antenna ratio is very high, then there may be signal loss and propagation delay occurs [21].

V. CONCLUSION

This paper contains different indoor localization techniques (AOA, TOF, TDOA, RSSI) and technologies (Wifi, UWB, RFID, Zigbee etc.). UWB Technology uses a sub-nano second radio pulse to transmit the data in a wide range of bandwidth. The main advantage of UWB based localization is its immune to multipath problem. Now a days the UWB based Real time Locating systems are used as tracking management system to acquire the attention in Industrial and logistic applications.

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